

# JIASHEN REN

Ph.D. Student in Data Science, City University of Hong Kong

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## EDUCATION

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**City University of Hong Kong**

Aug. 2026 – Present

Ph.D. in Data Science

Advisor: Prof. Xiangyu Zhao; Adaptive Machine Learning (AML) Lab

**Zhejiang University**

Sep. 2022 – May 2026

B.S. in Electrical Engineering

**University of Illinois at Urbana-Champaign**

Sep. 2022 – May 2026

B.S. in Electrical Engineering

## RESEARCH INTERESTS

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Human-centered and trustworthy AI; LLM behavior, evaluation, and security; constraint-aware machine learning.

## PUBLICATIONS

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- [1] Haoyu Zhu, Yao Zhang, **Jiashen Ren**, and Qingchun Hou. **T-SKM-Net: Trainable Neural Network Framework for Linear Constraint Satisfaction via Sampling Kaczmarz-Motzkin Method**. *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, 2026.

## RESEARCH IN PROGRESS

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### **Construct-Surgicity of Persona Prompts in LLM User Simulation**

- Studying whether persona prompts activate intended traits without shifting construct-distal preferences in LLM-based user simulation.
- Using trait-by-domain audits, null controls, and factorization tests to distinguish target-trait activation from cross-domain leakage across API-served and open-weight models.
- Examining how structured discriminant evidence and task design attenuate leakage while retaining target traits, with downstream evaluation in synthetic surveys and A/B choices.

## RESEARCH EXPERIENCE

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### **Constraint-Oriented Neural Networks for Power-System Optimization**

#### **SRTP Research Project**

Feb. 2025 – Aug. 2026

Supervisor: Prof. Qingchun Hou, ZJU-UIUC Institute, Zhejiang University

- Designed and implemented a physics-constrained neural network architecture for linear constraint satisfaction in power-system optimization.
- Worked on graph neural network-based grid decomposition and boundary-flow prediction, with validation on IEEE test systems for AC optimal power flow.

## RESEARCH EXPERIENCE (CONTINUED)

**DiffSinger-Based Voice Enhancement and Synthesis****Summer Research**

Jun. 2023 – Aug. 2023

*Supervisor: Prof. Zhou Zhao, College of Computer Science, Zhejiang University*

- Reproduced DiffSinger components, including noise scheduling and denoising networks, and studied the noise-to-Mel-spectrogram generation pipeline.
- Assisted with experimental analysis and manual annotation and evaluation of synthesized audio quality.

## SELECTED TECHNICAL PROJECT

**FPGA-Based Handwritten Digit Recognition**

Code

*Course project (ECE385)*

- Implemented a 1,024–32–10 multilayer perceptron in SystemVerilog, converting a PyTorch model to Q7.24 fixed-point arithmetic with custom multiply-accumulate units.
- Integrated camera input, image processing, VGA display, and I<sup>2</sup>C communication; achieved timing closure at 50 MHz.

## TEACHING EXPERIENCE

**Calculus III (MATH241)**

Sep. 2025 – Dec. 2025

*Teaching Assistant***Introduction to Differential Equations (MATH285)**

Jan. 2025 – May 2025

*Teaching Assistant*

## SELECTED HONORS AND AWARDS

**Honorable Mention**, Mathematical Contest in Modeling (MCM) 2024**National Silver Medal**, China International College Students' Innovation Competition 2023

## TECHNICAL SKILLS

**Programming:** Python, C, SystemVerilog.**Machine learning:** PyTorch, PyTorch Lightning, NumPy, pandas, scikit-learn.**Tools:** Git, Linux, L<sup>A</sup>T<sub>E</sub>X, Intel Quartus.**Languages:** Mandarin (native); English (proficient).